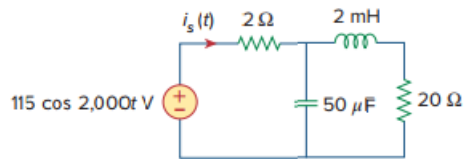


Problem

In the circuit of Fig. 9.54, determine the value of $i_s(t)$.

Figure 9.54:



Step-by-step solution

Step 1 of 6

Refer to Figure 9.54 in the text book.

Consider the value of the voltage source $v(t)$.

$$v(t) = 115 \cos 2,000t \text{ V}$$

Represent in phasor form.

$$\mathbf{V} = 115 \angle 0^\circ \text{ V}$$

Therefore, the value of the source in phasor form is, $115 \angle 0^\circ \text{ V}$.

Step 2 of 6

Calculate the value of the reactance due to inductance.

$$X_L = j\omega L$$

Substitute $2,000 \text{ rad/s}$ for ω and 2 mH for L .

$$\begin{aligned} X_L &= j(2,000)(2 \text{ m}) \\ &= j4 \\ &= 4 \angle 90^\circ \Omega \end{aligned}$$

Therefore, the value of the reactance X_L is, $4 \angle 90^\circ \Omega$.

Step 3 of 6

Calculate the value of the reactance due to capacitance.

$$X_C = \frac{1}{j\omega C}$$

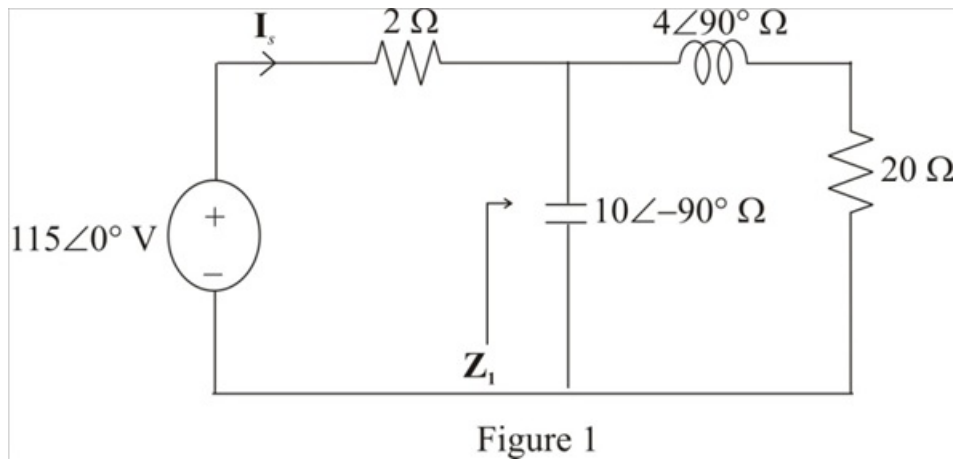
Substitute $2,000 \text{ rad/s}$ for ω and $50 \mu\text{F}$ for C .

$$\begin{aligned} X_C &= \frac{1}{j(2,000)(50 \mu)} \\ &= \frac{1}{j100 \text{ m}} \\ &= 10 \angle -90^\circ \Omega \end{aligned}$$

Therefore, the value of the reactance X_C is, $10 \angle -90^\circ \Omega$.

Step 4 of 6

Draw the circuit diagram in phasor representation.



Step 5 of 6

Calculate the value of the impedance Z_1 .

$$\begin{aligned} Z_1 &= (10\angle -90^\circ) \parallel (20 + 4\angle 90^\circ) \\ &= (10\angle -90^\circ) \parallel (20.4\angle 11.31^\circ) \\ &= \frac{(10\angle -90^\circ)(20.4\angle 11.31^\circ)}{(10\angle -90^\circ) + (20.4\angle 11.31^\circ)} \\ &= \frac{204\angle -78.69^\circ}{20.88\angle -16.69^\circ} \end{aligned}$$

$$Z_1 = 9.77\angle -62^\circ \Omega$$

Therefore, the value of the impedance Z_1 is, $9.77\angle -62^\circ \Omega$.

[Comments \(1\)](#)



Anonymous

How do you get to the answer $20.88\angle -16.69^\circ$??

Step 6 of 6

Calculate the value of the current I_s .

$$I_s = \frac{115\angle 0^\circ}{Z_1 + 2}$$

Substitute $9.77\angle -62^\circ \Omega$ for Z_1 .

$$\begin{aligned} I_s &= \frac{115\angle 0^\circ}{9.77\angle -62^\circ + 2} \\ &= \frac{115\angle 0^\circ}{10.85\angle -52.64^\circ} \\ &= 10.599\angle 52.64^\circ \text{ A} \end{aligned}$$

Represent in time domain.

$$i_s(t) = 10.599 \cos(2,000t + 52.64^\circ) \text{ A}$$

Therefore, the value of the current $i_s(t)$ is, $10.599 \cos(2,000t + 52.64^\circ) \text{ A}$.